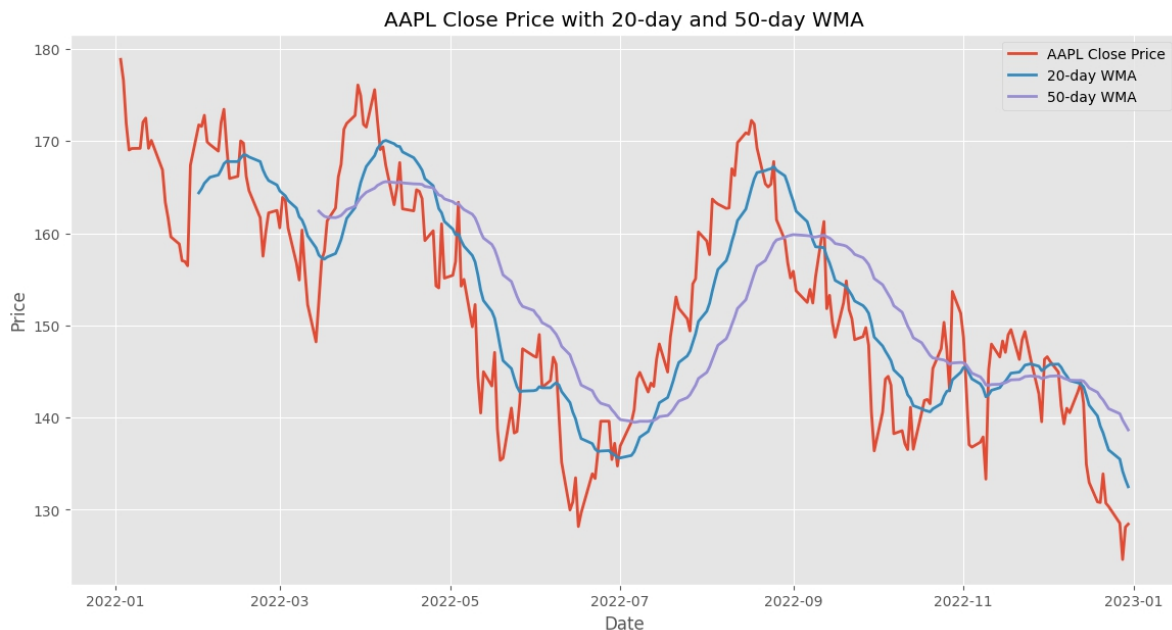




Each type of moving average has its proponents, and many trading systems use a combination of different averages to generate signals.

Relative Strength Index (RSI): Measuring Momentum

Developed by J. Welles Wilder in 1978, the Relative Strength Index is one of the most popular momentum oscillators. The RSI measures the speed and change of price movements on a scale from 0 to 100, helping traders identify overbought or oversold conditions.

Think of the RSI as a car's speedometer. Just as a speedometer tells you if a vehicle is moving too fast and might need to slow down, the RSI indicates when a market is moving up too quickly (overbought) and might be due for a pullback, or when it's falling too rapidly (oversold) and might be due for a bounce.

```
1. def calculate_rsi(data, window=14):  
2.     # Calculate price changes  
3.     delta = data.diff()  
4.  
5.     # Separate gains and losses  
6.     gain = delta.where(delta > 0, 0)  
7.     loss = -delta.where(delta < 0, 0)
```